$S1 == S2$

SEPP

Outcome of the Design process

1. Different modules required
2. Control relationships among modules
3. Interfaces among different modules
4. Data structures of the individual module
5. Algorithms required to implement the individual module

Module : collection of function & data

Design activities → Preliminary (high level)
→ Detailed design

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High-level : decomposed into set of modules
control rel^n & interfaces are identified
↳ outcome : program structure / software architecture

- Procedural development - diagram called structure chart
object - oriented design - UML

- In Detailed design - design datastructures & Algorithms
↳ outcome : module specification (MSPEC)

→ Design Methodology → procedural
→ object - oriented

→ Analysis vs Design activities differ in goal & scope

Analysis - goal : elaborate the customer req^n and does not consider implementation
↳ model is documented using graphical formalism

	function oriented	object oriented
analysis model	data flow diagrams	UML
design	structure chart	

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Characteristics of good s/w design:

1. Correctness.
2. Understandability
3. Efficiency - resource, time, cost optimisation issues
4. Maintainability - easy to change

If not,
- it lead to increased development cost
- efforts to maintain the product is more
- lead to program that is full of bugs and unreliable

An understandable design is modular & layered
- it should have foll^n characteristic to be easily understandable
1. should assign meaningful names to design components
2. should use decomposition & abstraction (divide & conquer principle)

Modularity - problem has been decomposed into small modules and have limited interactions

highly modular - high cohesion, low coupling] increased productivity

~~feature is~~ called functional independence.

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→ layered design - modules are arranged ~~int~~ in a hierarchy of layers.

- can be considered to be implementing control abstraction.
- simplifies debugging

→ Cohesion - measure of the functional strength of a module

Coupling - measure of the degree of interaction

↳ highly coupled ① function call involve passing large chunks of data

② Interactions occur through some shared data

Cohesion - when the funⁿ of the module ~~coopse~~ cooperate with each other for performing a single objective.

→ functional independence have follⁿ advantages:

① Error isolation - reduces the chances to the error propogating & easy to locate the error

② Scope of reuse - Reuse for the devlopment of other appⁿs.

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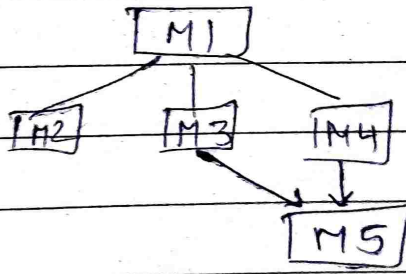
(5) Understandability

→ Important terminologies of layered design:
(1) Superordinate & subordinate modules
→ module that controls another module
↳ module controlled by another module

(2) Visibility - immediately lower layer modules are said to be visible.

(3) Control abstraction - modules at the higher level should not be visible to the modules at the lower layers. this is referred to as control --

(4) Depth & width



depth = 3
width = 3

(5) fan out - measure of no. of modules that are directly controlled by a given module
- high fan-out is not good. (M1 fanout = 3)

(6) fan in - no. of modules that directly invoke a given module. High fanin is good.
(M1 = 0, M2 = 1, M5 = 2)